TECHNICAL REPORT

Vertical Warehouse Sorting Station PLC-Based Industrial Automation

Elevator control, multi-floor conveyor sorting and computer-vision quality inspection, engineered as a hierarchical GRAFCET / GEMMA control architecture.

CODESYS V3.5 · Modbus TCP/IP · Factory I/O · Computer Vision Quality Control · HMI / WebVisualization

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Industrial automation project — PLC programming, safety logic, and HMI supervision

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1. Introduction

This project addresses the automation of a vertical goods-sorting station, designed to simulate an industrial warehouse distributed across several levels. The core of the system is a control logic programmed on a PLC (Programmable Logic Controller), which coordinates a main elevator, conveyor belts on different floors, and a quality-inspection system based on computer vision.

The implementation simulates a flexible production process that operates on demand, integrating the control logic with a supervision interface (HMI). This report covers the complete design of the automation system, from the technical specification, inputs and outputs, through to its programming and full operation.

2. Process Description

The operation of the automated station has been designed as a sequential state system, fully managed by the PLC. The process is divided into clearly differentiated phases, from order configuration to order fulfillment.

2.1 Initial Conditions and Preparation

Before any movement begins, the system verifies that the safety and availability conditions are met:

- The system must be powered and free of any active Emergency Stop.
- If a previous emergency occurred, the Reset sequence must have been executed to confirm the area has been cleared.
- The mode selector must be in the Automatic position.

2.2 Cycle Configuration and Start-Up

The process begins with the operator interacting through the HMI panel:

- **Data entry:** the operator enters the number of pieces requested for the current order.
- **Start:** pressing the START button validates the input and activates the green signal light.

2.3 Load-Unit Management (Ground Floor)

The first action of the automation system is to prepare the container in which the pieces will be placed:

- If no pallet is present in the elevator's loading zone, the system activates the Ground Floor emitter to generate an empty pallet.
- The infeed conveyors carry the pallet into the elevator.
- Once the position sensor confirms that the pallet is correctly positioned and centered, the elevator is ready to begin collecting the order.

2.4 Collection and Sorting Sequence

The elevator travels to the relevant upper floors to collect the requested material. On each floor, operation follows these steps:

- **Detection:** when a piece reaches the vision zone, the sensor captures its color value.
- **Logic decision:**
  - *Defective piece (black):* if the sensor detects the value corresponding to black, the PLC immediately reverses the conveyor. Simultaneously, the Defect Light on the HMI panel turns on to visually flag the rejection. This piece is sent down the reject chute and is not counted toward the order.
  - *Valid piece (color):* if the sensor detects a valid color, the system allows it to proceed. The floor conveyor and the elevator conveyor are synchronized to transfer the piece onto the pallet.
- **Counting:** every time a valid piece enters the elevator, the internal order counter is incremented by one.

2.5 Order Completion

The system continuously evaluates the completion condition: *pieces collected = pieces requested*.

- As soon as the target quantity is reached, the system stops the collection process.
- The elevator automatically descends to the Ground Floor.

- The output rollers activate to push the loaded pallet out toward the final dispatch area.
- The machine enters Stand-by, ready to receive a new order from the operator.

2.6 Interruption Handling

- **Pause (Stop):** if STOP is pressed, the machine halts its movements in a controlled manner, memorizing the current state so the cycle can be resumed later.
- **Emergency:** if the emergency mushroom button is activated, power to the actuators is cut immediately (hard stop), and the memorized cycle state is lost for safety reasons, requiring a manual system reset via the reset button. Once reset has been pressed and start is pressed again, the system resumes the order — but it is assumed that an operator has removed the boxes that had already left the floors. If this is not the case, the order will end up with one extra box.

2.7 Demonstration Video

A full walkthrough of the system in operation is available here: Demo video

3. Technical Specifications

The automated system must meet the following technical, functional, and safety requirements to guarantee correct order management and goods sorting.

3.1 Hardware and Field-Device Requirements

The control system manages an industrial plant simulated with Factory I/O, made up of the following physical subsystems:

- **Vertical elevation system:** an automated lift capable of positioning at four height levels — Ground Floor (receiving/dispatch) and Floors 1, 2, and 3 (processing).
- **Conveying systems:** independent motorized conveyors handling pallet infeed, elevator loading, and processing on each of the upper floors.
- **Vision instrumentation:** computer-vision sensors located at the entrance of Floors 1, 2, and 3. They output an integer value corresponding to the color of the piece for automatic classification.
- **Sorting actuators:** diverting mechanisms for the physical separation of defective material.

3.2 Control and Communication Requirements

- **Control unit:** the logic core is based on a Programmable Logic Controller (PLC) running in the CODESYS V3.5 environment.
- **Communication protocol:** interaction between the control logic and the physical plant is carried out over Modbus TCP/IP, guaranteeing real-time bidirectional data exchange (sensor inputs and actuator outputs).
- **Human-Machine Interface (HMI):** an operator panel is required to supervise plant status, manage alarms, and enter order data.

3.3 Functional Process Specifications

The automatic cycle must comply with the following operating rules:

- **Production on demand:** the system does not run indefinitely; it processes the number of pieces requested by the user, and the cycle ends automatically once that quantity is reached.
- **Quality control (rejection criterion):** pieces identified as “black” (the defect value) must be ejected from the line immediately by the reject actuators, accompanied by a dedicated visual signal (Defect Light) that stays active only during the rejection maneuver.
- **Valid-piece handling:** colored pieces are accepted and conveyed to the final container.

3.4 Safety Specifications and Operating Modes

- **Emergency priority:** activating the emergency mushroom button must immediately disconnect every power output, stopping motors and conveyors regardless of the process state.
- **Reset protocol:** recovery after an emergency is never automatic — an explicit manual confirmation sequence via the Reset pushbutton is required to validate that the area is safe before a new cycle is allowed.
- **Dual operating modes:** the system provides a physical or virtual selector to switch between:
  - **Manual mode:** direct control of actuators.
  - **Automatic mode:** execution of the automatic production logic sequence.

4. I/O Mapping

4.1 Digital Inputs

Address	Name	Description
%IX20.0	At_1	Presence sensor, Floor 1
%IX20.1	At_2	Presence sensor, Floor 2
%IX20.2	At_3	Presence sensor, Floor 3
%IX20.3	At_elevator	Piece-in-cabin detector
%IX20.4	At_entry	Presence sensor, infeed conveyor (Ground Floor)
%IX20.5	At_exit	Presence sensor, outfeed conveyor (Ground Floor)
%IX20.6	At_0_low	Low-position sensor, Ground Floor
%IX20.7	At_0_high	High-position sensor, Ground Floor
%IX21.0	At_1_low	Approach sensor, Floor 1
%IX21.1	At_1_high	Final-position sensor, Floor 1
%IX21.2	At_2_low	Approach sensor, Floor 2
%IX21.3	At_2_high	Final-position sensor, Floor 2
%IX21.4	At_3_low	Approach sensor, Floor 3
%IX21.5	At_3_high	Final-position sensor, Floor 3
%IX21.6	Start_Button	Physical cycle-start pushbutton
%IX21.7	Reset_Button	Physical system-reset pushbutton
%IX22.0	Stop_Button	Physical controlled-stop pushbutton
%IX22.1	Emergency_Stop	Emergency mushroom-button signal (negated logic)
%IX22.2	Auto	Selector in Automatic position
%IX22.4	Manual	Selector in Manual position

4.2 Digital Outputs

Address	Name	Description
%QX20.0	Conveyor_0	Ground-floor infeed conveyor motor
%QX20.1	Belt_Conveyor_2m_1_plus	Floor 1 conveyor — forward motor
%QX20.2	Belt_Conveyor_2m_2_plus	Floor 2 conveyor — forward motor
%QX20.3	Belt_Conveyor_2m_3_plus	Floor 3 conveyor — forward motor
%QX20.4	Exit_conveyor	Ground-floor outfeed conveyor motor
%QX20.5	Load	Elevator loading rollers
%QX20.6	Unload	Elevator unloading rollers
%QX20.7	Up	Elevation motor — up
%QX21.0	Down	Elevation motor — down
%QX21.1	Slow	Low-speed positioning mode
%QX21.2	Warning_light	Amber emergency/warning light
%QX21.3	Start_light	Green running indicator
%QX21.4	Fault_light	Defective-piece indicator
%QX21.5	Stop_light	Red stopped-system indicator
%QX23.2	Belt_Conveyor_2m_1_minus	Floor 1 conveyor — reverse (reject) motor
%QX23.3	Belt_Conveyor_2m_2_minus	Floor 2 conveyor — reverse (reject) motor
%QX23.4	Belt_Conveyor_2m_3_minus	Floor 3 conveyor — reverse (reject) motor
%QX23.5	Emitter_0	Pallet generator, Ground Floor
%QX23.6	Emitter_1	Box generator, Floor 1
%QX23.7	Emitter_2	Box generator, Floor 2
%QX24.0	Emitter_3	Box generator, Floor 3

4.3 Data Registers

Address	Name	Description
%IW0	Vision_Sensor_0_Value	Integer color value detected, Floor 1
%IW1	Vision_Sensor_1_Value	Integer color value detected, Floor 2
%IW2	Vision_Sensor_2_Value	Integer color value detected, Floor 3
%QW0	Display	General HMI display
%QW1	Display0	Floor 1 counter display
%QW2	Display2	Floor 2 counter display
%QW3	Display3	Floor 3 counter display
%QW4	Piezas_Defectuosas	Total accumulated rejected pieces

5. GEMMA — Operating and Stop States

GEMMA was used to represent the system's operating and stopped states, distinguishing automatic operation, manual operation, and the emergency stop. This diagram is the foundation for the rest of the automation logic.

5.1 Stop Procedures

- **A1 — Stop in the Initial State:** the absolute rest state of the automation system, where the machine remains powered but with no movement. The system waits for the operator to select automatic mode and press START to begin the production cycle.
- **A2 — Requested Stop at End of Cycle:** this transitional state is reached automatically when the current order's piece counter reaches zero. The machine completes any pending tasks — such as lowering the elevator to the ground floor and ejecting the pallet — before returning to rest in A1.
- **A5 — Resumption After Failure:** activated once the physical emergency button or the HMI virtual command has been released, indicating the area is safe again. In this phase, the system remains preventively locked, awaiting human confirmation (RESET) validating the integrity of the plant.
- **A6 — Return to the Initial State:** the logical reset procedure, during which the PLC clears memory and returns the actuators to their home configuration. Once all variables are zeroed and the mechanics are at rest, the system evolves back to the

waiting state A1.

5.2 Operating Procedures

- **F1 — Normal Production:** the operational core of the system, executing the automatic sequence of transport, computer-vision sorting, and palletizing. In this state, the PLC autonomously manages every movement based on the previously entered order.
- **F4 — Manual Mode:** allows individual control of actuators through the HMI interface, bypassing the programmed automatic logic sequence. This state is essential for preventive maintenance tasks, quick mechanical adjustments, or manually clearing obstructions on the line.

5.3 Failure Procedures

- **D1 — Emergency Stop:** the highest-priority state, causing the immediate disconnection of all power outputs upon activation of the physical button or an HMI command. The system remains in this critical state, blocking all movement, until the safety conditions required to begin the reset sequence are met.

6. GRAFCET Architecture

The control logic implemented in CODESYS uses a hierarchical GRAFCET architecture. This structure separates safety, mode management, and the production sequence into independent levels, making debugging easier and guaranteeing the robustness of the system.

- **Level 1 — Safety (GE):** supervises overall safety with absolute priority over the rest of the system. On an emergency, it activates the “Full Stop” state, stopping the actuators and requiring a manual reset sequence to restore service.
- **Level 2 — Operating Modes (GM):** manages system operability based on the physical selector’s state. It enables the production GRAFCET in “Automatic” or hands control over to the `POU_Manual` block for direct maintenance of actuators from the HMI.
- **Level 3 — Production (GP):** executes the production logic sequence (elevation, vision-based sorting, and palletizing) subordinate to Automatic mode. It implements an optimized structure with conditional jumps to collect material only from the floors requested by the order.

6.1 Emergency GRAFCET

This diagram represents the Emergency GRAFCET (GE), responsible for supervising the overall safety of the system with absolute priority over every other process. Its function is to continuously monitor the safety devices in order to force an immediate stop in the presence of any risk, ensuring that service is only restored under controlled conditions validated by the operator.

Initial stage — INIT. The process begins in the INIT stage, representing the system’s active-monitoring state. In this situation the machine is enabled to operate in its normal working modes, since no hazard conditions are present. This is the safety system’s rest state, in which automatic or manual cycles are allowed to run, remaining on standby until an anomaly is detected in the safety circuit.

Emergency transition. The system leaves the safe state immediately if the logic condition combining the physical mushroom button and the virtual HMI button is met. Negated logic is used for the physical emergency button, since it is a normally-closed contact — guaranteeing fail-safe behavior in the event of a broken cable. The transition triggers if either the physical button is pressed or the emergency command is activated from the HMI panel, giving priority to the stop regardless of which of the two signals fires.

EMERGENCY stage. Upon entering this stage, the PLC executes the highest-priority safety actions to protect the installation. The `Full Stop` variable is activated, which performs a complete software-level shutdown of all power outputs and stops the motors. At the same time, the `Warning_light` indicator turns on to visually warn floor personnel that the machine is stopped due to an active alarm state.

Reset and return-to-normal transition. Returning to the initial state requires more than simply releasing the emergency buttons — an explicit confirmation is required. The logic requires that the physical button be unlatched, the HMI button be deactivated, and that the operator press the Reset button, ensuring the area is hazard-free and has been verified.

6.2 Operating-Modes GRAFCET

This diagram corresponds to the Mode Management GRAFCET (GM), which acts as the intermediate level in the system’s control hierarchy. Its purpose is to interpret the physical selector’s position on the operator panel to exclusively enable either the automatic production cycle or manual maintenance control, ensuring the two modes can never overlap.

Stage 0 — Waiting for Selection. This is the initial rest stage, in which the system waits for the operator to choose a valid operating mode. While in this state, no motion control logic runs and no actuators may be driven. The flow leaves this stage only when the selector switches to a defined position, activating either the Auto or the Manual signal, which triggers the transition to the corresponding operating branch.

Stage 1 — Automatic Mode. Upon detecting the Auto signal, this stage is activated, whose function is to enable execution of the Production GRAFCET (GP). During this state, plant control is handed over to the PLC's autonomous logic sequence in order to fulfil orders. This stage remains active while the selector signal persists, but the moment the condition NOT Auto is detected, the system forces an immediate return to stage 0, halting the automatic sequence.

Stage 2 — Manual Mode. If the selection corresponds to the Manual signal, the system transitions to this stage, which calls the `POU_Manual` program block. Here the sequential logic is bypassed, and the operator is allowed to individually drive motors and conveyors from the HMI for adjustments or unblocking. As in the previous case, loss of the selection signal, detected as NOT Manual, causes the system to leave this state and return to the initial waiting stage.

6.3 Production GRAFCET

This diagram represents the Production GRAFCET (GP), which executes the main transport and sorting sequence. Its structure combines a linear vertical-movement sequence with decision loops on each floor to manage quality and piece count.

- **Stages 10–11 — Initialization and infeed management:** the cycle begins by establishing safety conditions, resetting the warning lights, and ensuring the loading motors and conveyors are stopped. The system then waits for the start signal and verifies that the elevator is empty and positioned at the ground floor.
- **Stage 12:** the requested-piece counter for the first floor is checked. If demand is zero, the sequence jumps directly to stage 29, skipping the entire Floor 1 block; if the counter is positive, the transition to begin collection is enabled.
- **Stage 13:** the `SUBIR_PISO1` output is activated, driving the elevator motor upward. This action is held until the position sensor detects that the cabin has reached the exact height of Floor 1.
- **Stage 14:** once the elevator has stopped, the up command is reset and the piece emitter and infeed conveyor are activated simultaneously, allowing material to enter the inspection zone to be captured by the vision system.
- **Stage 15:** this stage acts as a decision point based on the vision sensor's reading. The PLC evaluates the received integer value and branches to the rejection routine if the value corresponds to black, or to the acceptance routine if the box is a valid color.
- **Stage 16:** if a defective piece is detected, this stage reverses the conveyor's direction and activates the eject piston to push the material off the production line.
- **Stage 17:** a safety timer (`TON_ELIMINAR`) keeps the rejection actuators active long enough to guarantee the piece has physically left the line before requesting a new piece back in stage 13.
- **Stage 18:** if the piece is valid, the loading rollers are activated to bring it into the elevator, and the Floor 1 pending-piece counter is decremented by one.
- **Stage 19:** a safety wait confirms the correct amount of load inside the elevator before evaluating whether more pieces are needed from this floor or whether the sequence can move on.
- **Stage 20:** marks the end of operations on Floor 1. The corresponding variables are reset, enabling the transition to the evaluation of the next floor.

Stages 21–29 operate equivalently to Floor 1, the only difference being that this level handles green boxes. The process starts by checking whether the order requires this type of box; if so, the elevator rises and the camera inspects the material. If the piece is valid (green), it is loaded onto the pallet and subtracted from the counter; if a black box is detected, the system automatically ejects it by reversing the conveyor, repeating the cycle until the required units for this floor are complete.

- **Stage 30:** marks the end of collection across the different floors. The system executes a brief stabilization pause and resets the upper-floor position memory, preparing the elevator for its return trip to the delivery area.
- **Stage 31:** the down motor (`DOWN`) is activated to move the cabin from the third floor down to ground level. Movement continues until the ground-floor sensor detects the elevator's arrival, at which point the descent stops.
- **Stage 32:** once in the exit position, the elevator rollers and the outfeed conveyor are driven to eject the finished pallet from the machine. The system holds this action until it confirms the load has exited, marking the current order as complete.

Floor 3 operating sequence. Stages 33–39 operate identically to the previous floors, managing the Blue product in this case. If the order includes this material, the elevator rises to the third floor and the camera validates each piece; valid pieces are loaded, decrementing the counter, while defective pieces are automatically rejected until the demand for this floor is fulfilled.

Note: when redrawing the GRAFCET for documentation purposes, stage numbers were renumbered for a cleaner, sequential order. However, this renumbering could not be replicated inside CODESYS itself, since the rename function was not working as expected.

7. CODESYS Implementation

Programming was carried out in the CODESYS V3.5 environment, using a virtual PLC connected to the Factory I/O simulated plant over Modbus TCP/IP. The system logic is split across several programs and blocks to keep the code easy to follow, cleanly separating safety, operating modes, and the automatic cycle.

7.1 Program Organization

The project is structured around the following elements:

- **GVL:** holds the global variables of the system, grouping inputs and outputs, emergency and reset signals, the mode selector, and the variables shared between the PLC logic and the HMI.
- **MODO_EMERGENCIA (PRG):** manages the emergency stop and the system reset sequence.
- **MODO_AUTO (PRG):** implements the automatic process sequence, organized into distinct steps controlling elevator movement, conveyors, vision-based sorting, and piece counting.
- **POU_MANUAL:** handles manual control of the actuators from the HMI when the system is in manual mode.

All of these programs run cyclically within the PLC's main task, guaranteeing a continuous response to any change in the inputs.

7.2 Global Variable List (GVL)

The GVL (Global Variable List) is the centralized data backbone of the PLC, allowing physical signals, operator commands, and control logic to share the same memory space without conflicts. To organize the more than 80 variables in the project, they have been grouped by function within the industrial process:

Group	Function	Key variables
Sensors (coils)	Position and state detection	`At_elevator`, `At_1_high`, `At_3_low`, `Stop_Button`
Actuators	Motion and power control	`Up`, `Down`, `Slow`, `Conveyor_0`, `Emitter_1`
Vision and data	Analog processing and counting	`Vision_Sensor_0_Value`, `piso1`, `Piezas_Defectuosas`
Signaling	Warning states and alarms	`Start_light`, `Warning_light`, `TON_Blink`
HMI interface	Manual control and setpoints	`Manual_Subir`, `Start_HMI`, `Manual_P1_Crear`

Below is the variable declaration block used in the project. `BOOL` types are used for binary signals and `INT` for counters and vision-sensor values.

```
{attribute 'qualified_only'}
VAR_GLOBAL

    // --- POSITION AND STATE SENSORS ---
    At_1, At_2, At_3 : BOOL;
    At_elevator, At_entry, At_exit : BOOL;
    At_0_low, At_0_high, At_1_low, At_1_high : BOOL;
    At_2_low, At_2_high, At_3_low, At_3_high : BOOL;
    Start_Button, Reset_Button, Stop_Button : BOOL;
    Emergency_Stop : BOOL := TRUE;
    Auto, Manual : BOOL;

    // --- ACTUATORS AND DIGITAL OUTPUTS ---
    Conveyor_0 : BOOL;
    Belt_Conveyor_2m_1_plus, Belt_Conveyor_2m_2_plus, Belt_Conveyor_2m_3_plus : BOOL;
    Load, Unload, Up, Down, Slow : BOOL;
    Warning_light, Start_light, Fault_light, Stop_light : BOOL;
    Emitter_0, Emitter_1, Emitter_2, Emitter_3 : BOOL;
    Belt_Conveyor_2m_1_minus, Belt_Conveyor_2m_2_minus, Belt_Conveyor_2m_3_minus : BOOL;

    // --- PROCESS AND VISION DATA ---
    Vision_Sensor_0_Value, Vision_Sensor_1_Value, Vision_Sensor_2_Value : INT;
    Display, Display0, Display2, Display3, Piezas_Defectuosas : INT;
    piso1, piso2, piso3 : INT; // Order-tracking auxiliary variables
    NEGRA : INT := 7;         // Constant for defect detection
    TON_Blink : TON;           // Timer for emergency-light blinking

    // --- HMI-ONLY VARIABLES (MANUAL MODE) ---
    Start_HMI, Stop_HMI, Reset_HMI, Emergencia_HMI : BOOL;
    Manual_Subir, Manual_Bajar, Manual_Cargar, Manual_Descargar : BOOL;
    Manual_P1_Crear, Manual_P1_Avanzar, Manual_P1_Retroceder : BOOL;
    // ... (same structure repeated for Manual_P2 and Manual_P3)
END_VAR
```

7.3 Emergency-Stop Management

System safety is handled by the `MODO_EMERGENCIA` program, configured with execution priority over the rest of the logic (automatic or manual cycles). This program implements a deterministic state diagram designed to guarantee fail-safe behavior.

Trigger and latch logic. The system continuously monitors the safety inputs (physical button and HMI button). When either is activated (`NOT Emergency_Stop` or `Emergencia_HMI`), the PLC immediately forces the transition to the Emergency state.

In this state, the PLC executes a Full Stop action that overrides any motion command, forcing every actuation variable to `FALSE` (logic 0). This guarantees the power is cut to motors and pneumatic actuators, killing the system's inertia.

```
// --- FORCING ACTUATORS TO SAFE STATE (OFF) ---

// 1. Elevator stop
GVL.Up := FALSE;
GVL.Down := FALSE;
GVL.Slow := FALSE;      // Disable speed modes
GVL.Load := FALSE;      // Lock loading rollers
GVL.Unload := FALSE;    // Lock unloading rollers

// 2. Ground-floor transport stop
GVL.Conveyor_0 := FALSE;
GVL.Exit_conveyor := FALSE;
GVL.Emitter_0 := FALSE; // Inhibit new material infeed

// 3. Floor actuator stop (1, 2 and 3)
// Forward, reverse and reject-piston conveyors are all cut
GVL.Belt_Conveyor_2m_1_plus := FALSE;
GVL.Belt_Conveyor_2m_1_minus := FALSE;
GVL.Remover_planta1 := FALSE;
// (Same logic applied to GVL.Emitter_x and the Floor 2/3 conveyors)
```

Signaling and visual warning. Simultaneously with the power cut, the system alerts operators via light signals. Normal-operation lights are switched off, and a blinking amber light (`Warning_light`) is activated. To achieve the blink effect without physical relays, an oscillator was programmed using a `TON` timer in the PLC:

```
// --- SIGNALING MANAGEMENT ---
GVL.Start_light := FALSE;
GVL.Stop_light := FALSE;

// Blink-frequency generation (500ms)
GVL.TON_Blink(IN := NOT GVL.TON_Blink.Q, PT := T#500MS);
GVL.Warning_light := GVL.TON_Blink.Q; // Warning light
```

Reset protocol. Leaving the emergency state requires a strict protocol: releasing the physical button and pressing the RESET button. During the exit transition (`_aInit_entry`), a data-cleanup routine runs to prevent the system from restarting with "ghost" values from previously interrupted cycles. Piece counters and HMI displays are reinitialized:

```
// --- DATA RE-INITIALIZATION ROUTINE ---
GVL.Display0 := 0;          // Reset Floor 1 counter
GVL.Display2 := 0;          // Reset Floor 2 counter
GVL.Display3 := 0;          // Reset Floor 3 counter
GVL.Display := 0;           // Reset general counter
GVL.Piezas_Defectuosas := 0; // Reset reject counter
```

7.4 Operating-Mode Management

The control architecture implements mode selection based on mutual exclusion, guaranteeing the system can never run under two contradictory logics at once. This is handled in the main program, which evaluates the physical selector and routes execution to the corresponding routine: `MODO_AUTO` or `POU_MANUAL`.

Manual-mode implementation (direct/jog control). The `POU_MANUAL` block was designed under a "direct command" (jog operation) logic. In this state, the automatic sequence is bypassed and a direct mapping is created between the HMI's virtual buttons and the PLC's physical outputs (GVL global variables).

This lets the operator drive motors and actuators individually to clear jams or verify the behavior of specific components. The direct-assignment logic implemented is as follows:

```
// --- MANUAL CONTROL LOGIC (HMI -> Actuator mapping) ---

// --- 1. ELEVATOR (CABIN) ---
GVL.Up     := GVL.Manual_Subir;
GVL.Down   := GVL.Manual_Bajar;
GVL.Load   := GVL.Manual_Cargar;
GVL.Unload := GVL.Manual_Descargar;

// --- 2. GROUND FLOOR (INFEED) ---
GVL.Conveyor_0 := GVL.Manual_Entrada_Activa;
GVL.Emitter_0  := GVL.Manual_Crear_Pale;

// --- 3. FLOOR 1 (BLUE) ---
GVL.Emitter_1 := GVL.Manual_P1_Crear;
GVL.Belt_Conveyor_2m_1_plus  := GVL.Manual_P1_Avanzar;
GVL.Belt_Conveyor_2m_1_minus := GVL.Manual_P1_Retroceder;

// --- 4. FLOOR 2 (GREEN) ---
GVL.Emitter_2 := GVL.Manual_P2_Crear;
GVL.Belt_Conveyor_2m_2_plus  := GVL.Manual_P2_Avanzar;
GVL.Belt_Conveyor_2m_2_minus := GVL.Manual_P2_Retroceder;

// --- 5. FLOOR 3 (BROWN) ---
GVL.Emitter_3 := GVL.Manual_P3_Crear;
GVL.Belt_Conveyor_2m_3_plus  := GVL.Manual_P3_Avanzar;
GVL.Belt_Conveyor_2m_3_minus := GVL.Manual_P3_Retroceder;
```

7.5 Automatic-Cycle Implementation

The main process logic is encapsulated in the `MODO_AUTO` (PRG) program unit. To keep the code modular and maintainable, a mixed architecture based on the IEC 61131-3 standard was chosen:

- **Sequential control (SFC):** the main flow (stages and transitions) is managed through a Sequential Function Chart (GRAFCET), allowing the process state to be visualized in real time.
- **Action logic (Ladder/ST):** the concrete operations that occur within each stage (moving the elevator, driving conveyors, processing vision data) are segregated into independent actions. As shown in the project structure, each critical maneuver has its own code block (e.g. `SUBIR_PISO1`, `RECOGER_OBJ`, `ELIMINAR_OBJ`).

7.5.1 Elevator motion logic. The elevator's positioning logic was implemented in Ladder to manage precise ascent. The system first checks safety conditions (load present in the elevator, no obstacles at the infeed) before activating the up motor (`GVL.Up`) and the position marker. An intermediate sensor (`GVL.At_1_low`) is also used to activate low speed (`GVL.Slow`) just before arrival, guaranteeing a smooth, level stop once the floor's final sensor (`GVL.At_1_high`) triggers. This same safety–approach–stop structure is replicated for the Floor 2 and Floor 3 maneuvers.

The descent maneuver first checks for the presence of the pallet in the elevator via the `GVL.At_elevator` sensor, to avoid descending empty, then resets the up and load flags for safety. Next, the `GVL.Down` coil is activated with a `SET` instruction, starting the vertical downward motion. During the trip, the `GVL.At_1_low` sensor activates low speed `GVL.Slow`, allowing a controlled deceleration before arrival.

7.5.2 Object creation and removal. Handling of the loads is implemented with combinational Ladder logic that coordinates position sensors with transport actuators.

The object-creation logic synchronizes the elevator's arrival with material infeed using `SET` instructions. The system simultaneously activates the emitter (`Emitter_X`) and the forward conveyor (`Belt_Conveyor...plus`) only when the floor sensor detects the cabin (`At_X_high`) and confirms the down motor is stopped (`/GVL.Down`). This maneuver ensures the piece is generated and enters the vision zone exactly when the elevator is stable and positioned for loading.

The removal routine uses `EQ` comparison blocks to detect defective pieces when the vision-sensor value matches the reject code (7). Once this condition is met on the active floor, the conveyor direction is reversed with a `SET` instruction on the `...minus` variable. Simultaneously, the `TON_ELIMINAR` timer keeps this ejection active for 3 seconds, guaranteeing the object fully leaves the line.

The pickup routine manages the physical transfer of validated material from the floor conveyor into the elevator. Once the piece is confirmed correct, the system simultaneously activates the loading rollers (`GVL.Load`) and keeps the conveyor moving forward, ensuring the correct transfer onto the pallet before updating the order counter.

7.5.3 Timers and counters. Two auxiliary timers guarantee load stability and correct operation. `T_Espera` introduces a brief stabilization pause before vertical movements, ensuring the load does not shift due to inertia. `TIEMPO_FINAL` manages the outfeed conveyor in the final stage, keeping it active for exactly as long as needed to guarantee the pallet fully leaves the area.

On one hand, valid-piece accounting updates the remaining order by subtracting one unit from the floor variable and immediately refreshing the value on the HMI display. It also performs housekeeping by forcing the timer inputs to `FALSE`, so they can count from zero again on the next iteration without getting stuck:

```
GVL.piso1 := GVL.piso1 - 1;
T_Espera(IN := FALSE);
TON_AVASTECIMIENTO(IN := FALSE);
TON_ELIMINAR(IN := FALSE);

GVL.Display0:=GVL.piso1;
```

On the other hand, when a defective piece is detected, the system increments the cumulative fault counter to keep track of quality. As in the acceptance routine, it is mandatory to reset the timer instances (`IN := FALSE`) to make sure the system clears any previous time memory and is ready to evaluate a new entry.

```
T_Espera(IN := FALSE);
TON_AVASTECIMIENTO(IN := FALSE);
TON_ELIMINAR(IN := FALSE);
GVL.Piezas_Defectuosas:=GVL.Piezas_Defectuosas +1;
```

7.6 Stops and System Behavior

During automatic operation, the system can be stopped in a controlled way via the STOP button, preserving the cycle state so it can be resumed later. An emergency stop, on the other hand, causes an immediate halt of every actuator and requires a manual reset before the machine can be restarted.

8. HMI

To enable remote monitoring, the WebVisualization service was enabled on port 8080, allowing any operator with a web browser to supervise the status of the pieces.

8.1 Main Panel

Element	Type	Description and function
START button	Pushbutton (NO)	Starts the automatic operating cycle if initial conditions are met
STOP button	Pushbutton (NC)	Stops the operating cycle in a controlled way (end-of-cycle stop or pause)
EMERGENCY button	Mushroom / pushbutton	Immediately stops every actuator in the system for safety
RESET button	Pushbutton (NO)	Resets the system after an emergency stop
MODE switch	Selector	Switches the machine between “Manual mode” and “Automatic mode”
START light (green)	Indicator	Signals the machine is running the automatic cycle
STOP light (red)	Indicator	Signals the machine is stopped or on standby
EMERGENCY light (amber)	Indicator	Warns that the emergency button has been pressed and the system is safety-locked
DEFECT light (black)	Indicator	Turns on temporarily when the system detects and rejects a defective (black) piece

8.2 Order Panel

Numeric displays (Floors 1–3) show the count of processed pieces or the current order status for each floor.

8.3 Manual Panel

Element	Description and function
UP / DOWN	Drive the elevator motor to move it vertically up or down
LOAD / UNLOAD	Activate the elevator’s internal conveyor to bring in or remove the load
CREATE PALLET	Forces the emission of a new empty pallet on the infeed conveyor
INFEED CONVEYOR	Activates the Ground Floor conveyor motor
CREATE BOX (blue/green)	Manually generates a piece of a specific color at the corresponding floor’s emitter, for testing
FORWARD (Floors 1, 2, 3)	Drives the corresponding floor’s conveyor forward
REVERSE (Floors 1, 2, 3)	Drives the corresponding floor’s conveyor in reverse

Demo video: Watch on Google Drive